Transport-scheme comparison — chain vs. SC vs. octree, with and without delta-M, ranked against a fine reference

Comparison of the three transport configurations of the `rt3d` solver on the MANUAL §11 worked example, each run with and without delta-M (Wiscombe) scaling, **plus a fine-octree reference solution and grid-refinement runs that rank every configuration by accuracy** (plan: `plan/compare-transport-plan.md` + `plan/transport-ranking-plan.md`); all figures and CSVs live in `plot/compare_transport/`. Measured 2026-10-14 on Windows with the `pyarts` interpreter; physics numbers are machine-independent, wall times are not (the previous version of this report was measured on macOS/arm64 — wall times differ by ~2–4× between the two machines; Ng-accelerated iteration counts can also shift by a few iterations with BLAS rounding, e.g. SC δ-off: 51 → 44).

Reproduce with:

```
$env:Path = "$env:USERPROFILE\miniforge3\envs\pyarts\Library\bin;" + $env:Path
& "$env:USERPROFILE\miniforge3\envs\pyarts\python.exe" scripts/compare_transport.py
```

(3D HTML scenes need `trame-pyvista`; installed in the Windows `pyarts` env on 2026-10-14, so all scenes and `vol_j.png` are current. Per-run solutions are cached in `plot/compare_transport/_solve_cache/` (gitignored) — reruns skip the solves.)

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1. Purpose and run matrix

The solver currently offers three transport algorithms and an optional delta-M split of the scattering operator. This study answers four questions on one physically non-trivial case (strongly forward-scattering cloud, solar beam, reflecting ground):

1. **chain vs. SC** — how large is the pure *transport-algorithm* error when both run on *identical cells*?
2. **unigrid vs. octree** — the grid-refinement error (validation anchor 5).
3. **delta-M on/off** — does it accelerate convergence, and does the converged solution stay the same (the equivalence claim of `plan/delta-m-plan.md`)?
4. **which run is best?** — pairwise differences alone cannot answer this (a gap between two approximations says nothing about which is closer to the exact solution), so the study adds a fine reference solution and ranks every run by its error against it (section 5).

Six comparison runs plus five ranking runs:

run key	grid	transport	delta-M	role
`uni_chain_dM0`	unigrid 24^3	lattice-stencil chains	off	comparison
`uni_chain_dM1`	unigrid 24^3	lattice-stencil chains	M = 1	comparison
`uni_sc_dM0`	unigrid 24^3	single-cell short characteristics	off	comparison

run key	grid	transport	delta-M	role
`uni_sc_dM1`	unigrid 24^3	single-cell short characteristics	M = 1	comparison
`octree_dM0`	octree (25 264 leaves)	exact characteristic marches	off	comparison
`octree_dM1`	octree (25 264 leaves)	exact characteristic marches	M = 1	comparison
`uni_chain48_dM1`	unigrid 48^3 (h = 0.208 m)	chains	M = 1	chain refinement
`uni_sc48_dM1`	unigrid 48^3	short characteristics	M = 1	SC refinement
`octree_d4_dM1`	octree τ_{max} = 1.0, depth 4 (34 476 leaves)	exact marches	M = 1	octree refinement
`octree_ref_dM1`	octree τ_{max} = 0.5, depth 5 (50 380 leaves)	exact marches	M = 1	fine-grid δ-M check
`octree_ref_dM0`	same octree as `octree_ref_dM1`	exact marches	off	**the reference**

---

2. The setups

2.1 Common physical case (MANUAL §11, unchanged)

Gaussian cloud + Lambertian ground + solar beam on the domain x, y ∈ [−5, 5], z ∈ [0, 10] m, periodic in x and y, space at the top, ground at the bottom. Units: lengths m, κ in m^{-1}, radiometric quantities normalised to the incident beam flux $F_0 = 1$.

- **Extinction** $\kappa_e = C \cdot \text{gauss}(\sigma = 1\text{ m}, \mu = (0, 0, 4)\text{ m})$, C scaled so the maximum vertical column optical depth is $\tau = 10$ (peak $\kappa_e \approx 3.9\ m^{-1}$, mean free path ≈ 0.26 m).
- **Scattering** $\kappa_s = \omega_{local} \cdot \kappa_e$ with ω_{local} = gauss/max(gauss), capped at $\omega \le 0.9$ (`omega_cap=0.9`) so plain lambda iteration stays tractable.
- **Asymmetry** g_{local} = gauss/(1.05·max) → peak g ≈ 0.952 in the cloud core (HG phase function) — the showcase regime for delta-M.
- **Thermal emission** B = 0 (pure scattering/beam problem).
- **Solar beam**: direct collimated beam, 30° from zenith, propagating in the x–z plane: $\omega_{sun} = [\sin 30°, 0, -\cos 30°]$, $F_0 = 1$.
- **Boundary conditions**: isotropic background I = 1 on the top face; Lambertian ground, albedo 0.3, emissivity 0 (no thermal emission).

2.2 Direction grid and iteration scheme (identical in all runs)

- Icosahedral direction grid, refinement 1 → **42 ordinates** (development grid), solid-angle weights from spherical-triangle areas, $\Sigma w = 4\pi$ (−1.8e−15).
- Lambda iteration with **1D Ng acceleration**, convergence on the relative source change, tol = 1e−6, cap 60 iterations.
- The 42-ordinate *angular* error is identical in every run (including the reference) and therefore cancels in all comparisons; quantifying it is the phase-6 direction-convergence study.

2.3 Grids

- **Unigrid 24³**: h = 0.4167 m, 13 824 cells; **unigrid 48³**: h = 0.2083 m, 110 592 cells.
- **Octree** base cell h_0 = 1.25 m, refinement driven by the *physical* κ_e, 2:1 balanced:
- depth 3 (τ_max = 2.0): **25 264 leaves** (L0: 316, L1: 788, L2: 3 680, L3: 20 480), cell size 1.25 m in the vacuum tails down to 0.156 m;
- depth 4 (τ_max = 1.0): **34 476 leaves** (finest cells still 0.156 m — the extra refinement acts on the intermediate levels);
- depth 5 (τ_max = 0.5): **50 380 leaves**, finest cells **0.078 m** (τ_cell ≤ 0.5 in the cloud core). κ_e-driven refinement *saturates* here: τ_max < 0.5 or depth > 5 adds no leaves, so this is the finest discretization the solver builds on this case — hence the reference.

2.4 Transport algorithms

- `chain` **(unigrid)** — lattice-stencil chain sweeps (`rt3d/characteristics.py`, DOIT-style): per ordinate one integer stencil s, characteristics run center-to-center across the whole grid in multi-cell chains. Exact for piecewise-constant sources along the chain; carries the known *grazing-ordinate stencil error* on coarse grids (MANUAL §6, anchor 5).
- `sc` **(unigrid)** — single-cell short characteristics (`rt3d/short_characteristics.py`, `plan/plan-sc-unigrid.md`): gap segment + target segment of equal length, bilinear 4-point face interpolation, wrap-around passes on periodic entry axes. Local and gather-only; more diffusive on coarse grids (path error ~ linear in cell size).
- **Octree** — exact characteristic marches (`transport.precompute_octree_march` + `sweep_direction_octree_exact`): one backward ray march per leaf per ordinate along the *true* ordinate direction, no lattice-stencil approximation. Short characteristics are *not available* on the octree (the solver raises `ValueError` for `transport="sc"` there).

2.5 delta-M scaling (`delta_M=1`, delta-Eddington)

Wiscombe split of the HG phase function, defined on the *row-normalized* discrete operator (MANUAL §9, `plan/delta-m-plan.md`):

```
Phi_delta = Phi_HG - (1/4pi) * sum_{n<1} (2n+1) g^n P_n(Omega.Omega')
kappa_e_delta = kappa_e - kappa_s * g^M        (diffuse transport only)
```

with the explicit `-omega·g^M·I/(1 - omega·g^M)` term in the source. Only the scattering part of the extinction is scaled; the direct beam keeps its physical attenuation, and octree refinement stays driven by the physical κ_e. delta-M changes the *iteration* (fewer iterations at g ≈ 0.95), not the converged discrete solution — the on/off comparison below verifies this.

2.6 The reference solution

`octree_ref_dM0`: the depth-5 octree (h_min = 0.078 m ≈ 3.3 cells per mean free path in the cloud core), exact marches, **delta-M off** so it carries no O(τ_cell) quadrature gap of the explicit −ω g^M I term. It is the finest discretization available on this case; its own remaining error is bounded by the depth-3 → depth-4 → depth-5 convergence sequence (section 5.2, error roughly halving per level) and by the fine-grid δ-on↔δ-off check (0.19% max, section 4.1).

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3. Diagnostics

All cell fields are sampled onto one **common 64^3 lattice** at the baseline octree cell size (h = 0.15625 m) — exact pointwise sampling for piecewise-constant fields, the finer reference cells are sampled, not averaged — so every pair of runs is directly pointwise comparable.

- **Mean intensity** $J = (1/4\pi)\int I\, d\Omega$.
- **Angular variance** $\sigma^2(r) = (1/4\pi)\int (I - J)^2\, d\Omega$ (`rt3d.diagnostics.angular_variance`) — zero for an isotropic field, measures the angular anisotropy of the radiation field.
- **Upward radiance at TOA**: the upward flux map $F^+(x, y) = \int_{\mu>0} I\, \mu\, d\Omega$ at the top layer, and the domain-mean upward radiance per upward ordinate (21 of the 42 ordinates have $\mu > 0$).

Six comparison pairs (B − A): SC↔chain per δ, δ-on↔δ-off per method, octree↔unigrid-chain per δ. Plus the **accuracy ranking**: every run against `octree_ref_dM0` (`ranking.csv`).

---

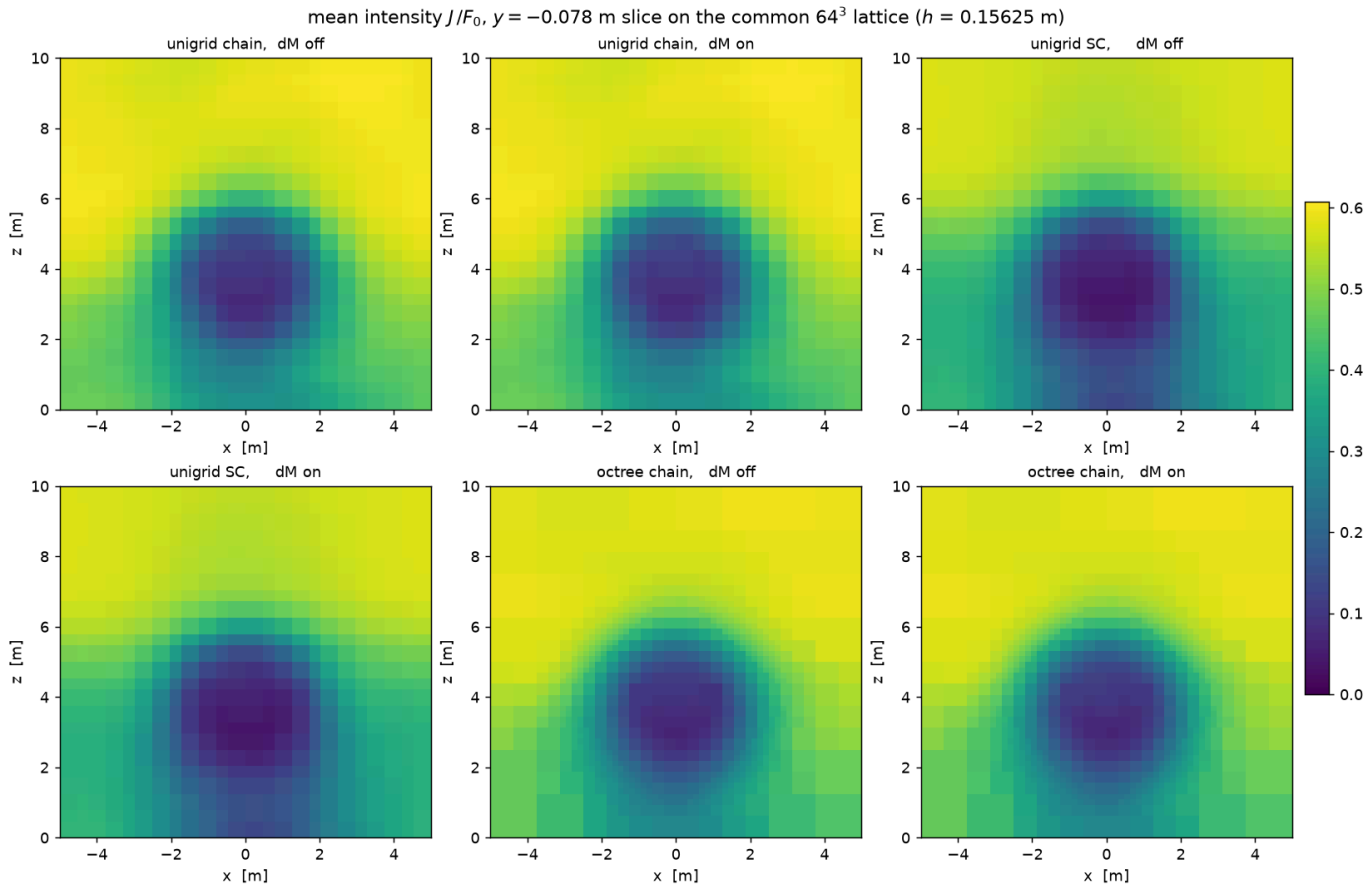
4. Results

4.1 Convergence and efficiency

run	transport	δ-M	n_iter	converged	wall [s]	⟨J⟩	mean F_z @TOA
`uni_chain_dM0`	chain	off	32	✓	3.4	0.5188	−2.5215
`uni_chain_dM1`	chain	M=1	**12**	✓	1.5	0.5188	−2.5214
`uni_sc_dM0`	sc	off	44	✓	24.7	0.4632	−2.5935
`uni_sc_dM1`	sc	M=1	**12**	✓	7.0	0.4633	−2.5933
`octree_dM0`	octree march	off	30	✓	95.3	0.4014	−2.5312
`octree_dM1`	octree march	M=1	**12**	✓	47.5	0.4016	−2.5308
`uni_chain48_dM1`	chain 48^3	M=1	13	✓	19.5	0.5194	−2.5223
`uni_sc48_dM1`	sc 48^3	M=1	12	✓	32.4	0.4640	−2.5927
`octree_d4_dM1`	octree march d4	M=1	12	✓	74.5	0.4114	−2.5319
`octree_ref_dM1`	octree march d5	M=1	12	✓	132.3	0.3841	−2.5319
`octree_ref_dM0`	octree march d5	off	27	✓	231.6	0.3840	−2.5320

- **delta-M is a large accelerator in this regime**: 32 → 12 (chain), 44 → 12 (SC), 30 → 12 (octree d3), 27 → 12 (reference grid) iterations; wall time drops by ~2× everywhere and by 3.5× for SC.
- The unigrid TOA mean F_z reproduces the MANUAL §11 numbers exactly (−2.5215 unigrid, −2.5312 octree d3).
- **δ-M equivalence at the finest grid**: `octree_ref_dM1` vs `octree_ref_dM0` differ by 0.19% max / 0.01% mean in J — half the depth-3 gap (1.3%), confirming the O(τ_cell) scaling of the explicit term and that the reference is δ-M-independent.

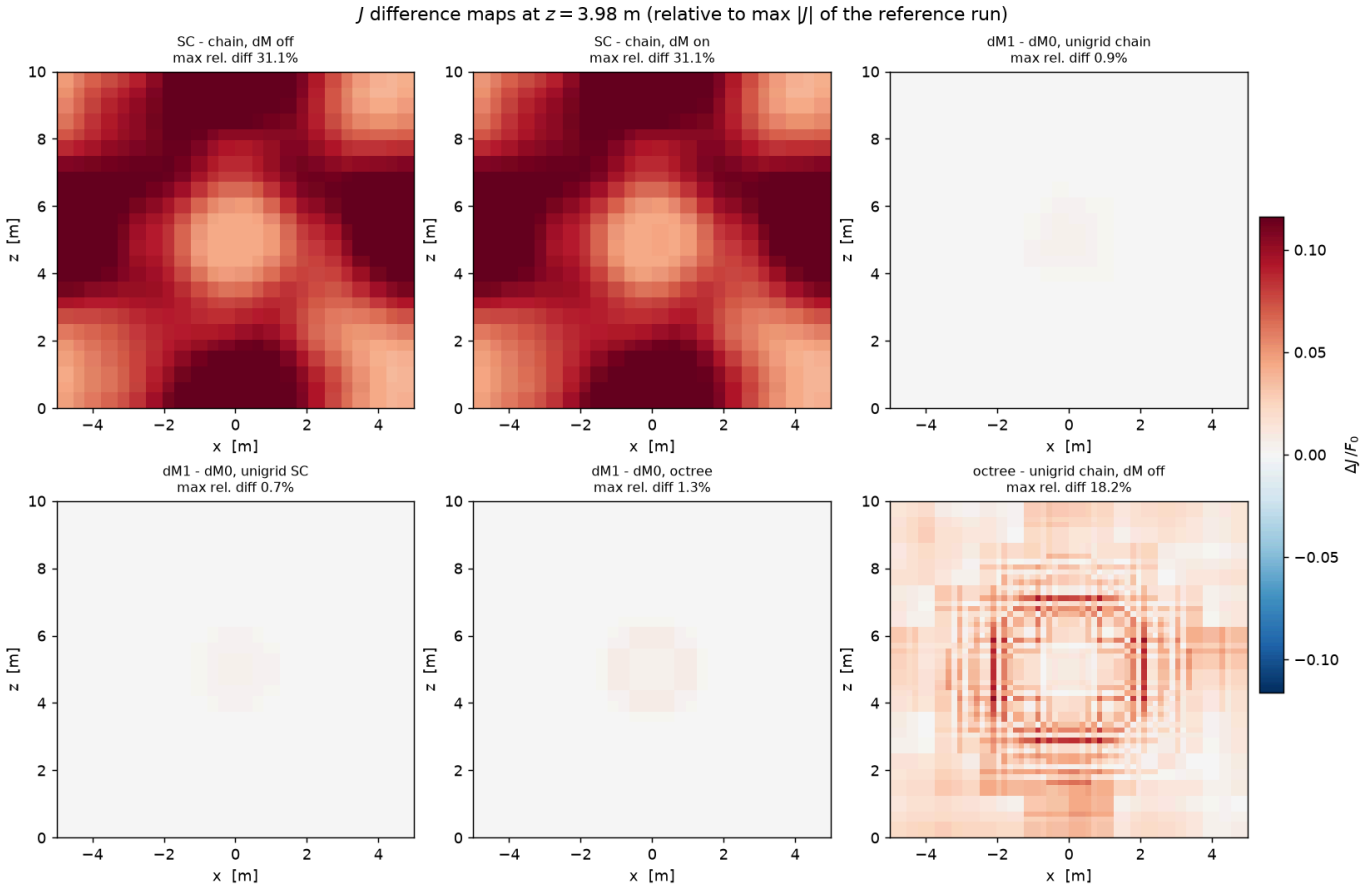
4.2 Mean intensity J



Band-mean J profiles (8 base-cell bands, `profiles.csv`; δ-on/off pairs agree to ≤3e−4, δ-off values shown):

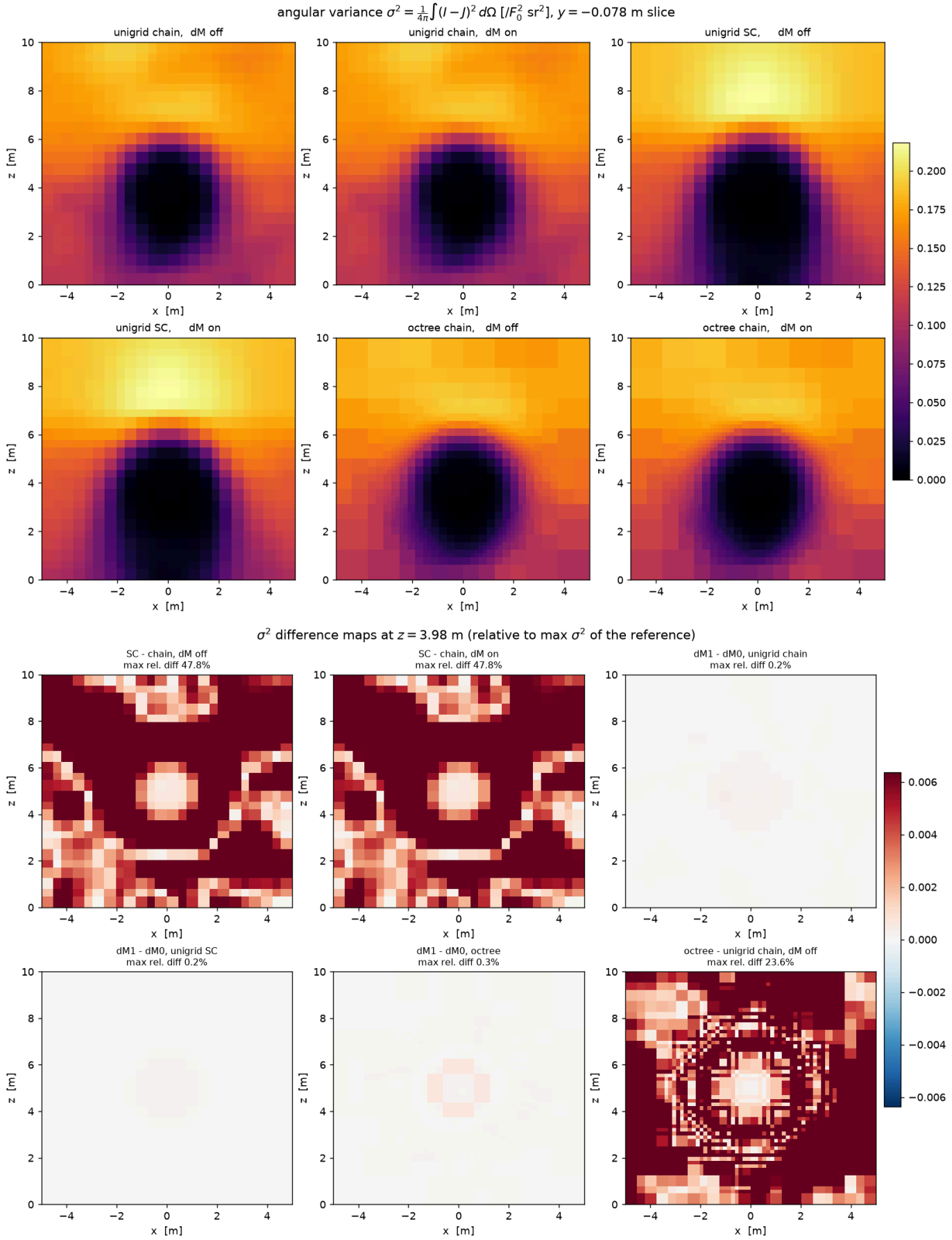
z [m]	chain 24^3	SC 24^3	octree d3	chain 48^3	SC 48^3	octree d4	reference
0.625	0.4372	0.3680	0.3095	0.4382	0.3660	0.3102	0.3192
3.125	0.4529	0.3629	0.3431	0.4539	0.3636	0.3587	0.3187
4.375	0.4964	0.4077	0.4154	0.4974	0.4105	0.4143	0.3296
6.875	0.5870	0.5631	0.5377	0.5872	0.5642	0.5443	0.5512
9.375	0.5893	0.5712	0.5850	0.5894	0.5709	0.5849	0.5849

Note the reference column: in the cloud core (z = 3.1–4.4 m) the reference J is far *lower* (0.32–0.33) than every depth-3-or-coarser run (0.41–0.50) — the coarse grids overestimate in-cloud radiation trapping; only the depth-4/5 octree resolution comes close.



- **chain vs. SC on identical cells is the largest pairwise discrepancy of the six-run study** (max 31%, mean 9%): SC's extra numerical diffusion lets more radiation penetrate the optically thick core, where it is absorbed (ω = 0.9, τ = 10) — J is lower everywhere below and inside the cloud (0.368 vs 0.437 in the bottom band).
- **unigrid-chain vs. octree** (18% max, 2.8% mean) reproduces the anchor-5 pattern of the manual: the unigrid grazing-ordinate stencil error inflates J below the cloud (0.437 vs 0.310 in the bottom band, the manual's 29%), with a ring structure around the cloud in the difference maps.
- **delta-M pairs are flat** (≤1.3% max, ~0.01–0.02% mean): the converged solution is independent of δ-M, as designed; the residual pointwise gap is the documented O(τ_cell) quadrature error of the explicit −ω g^M I term.

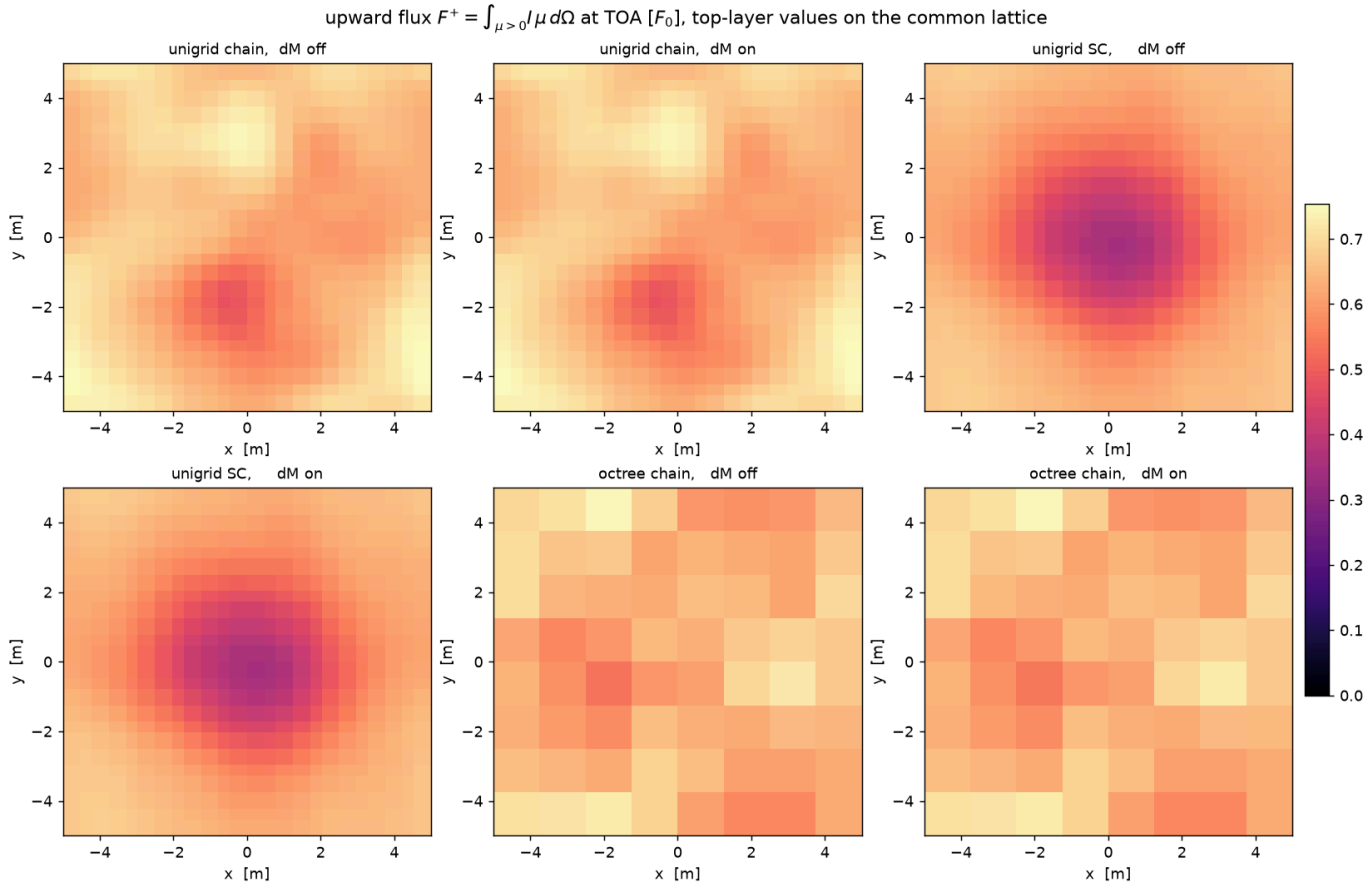
4.3 Angular variance

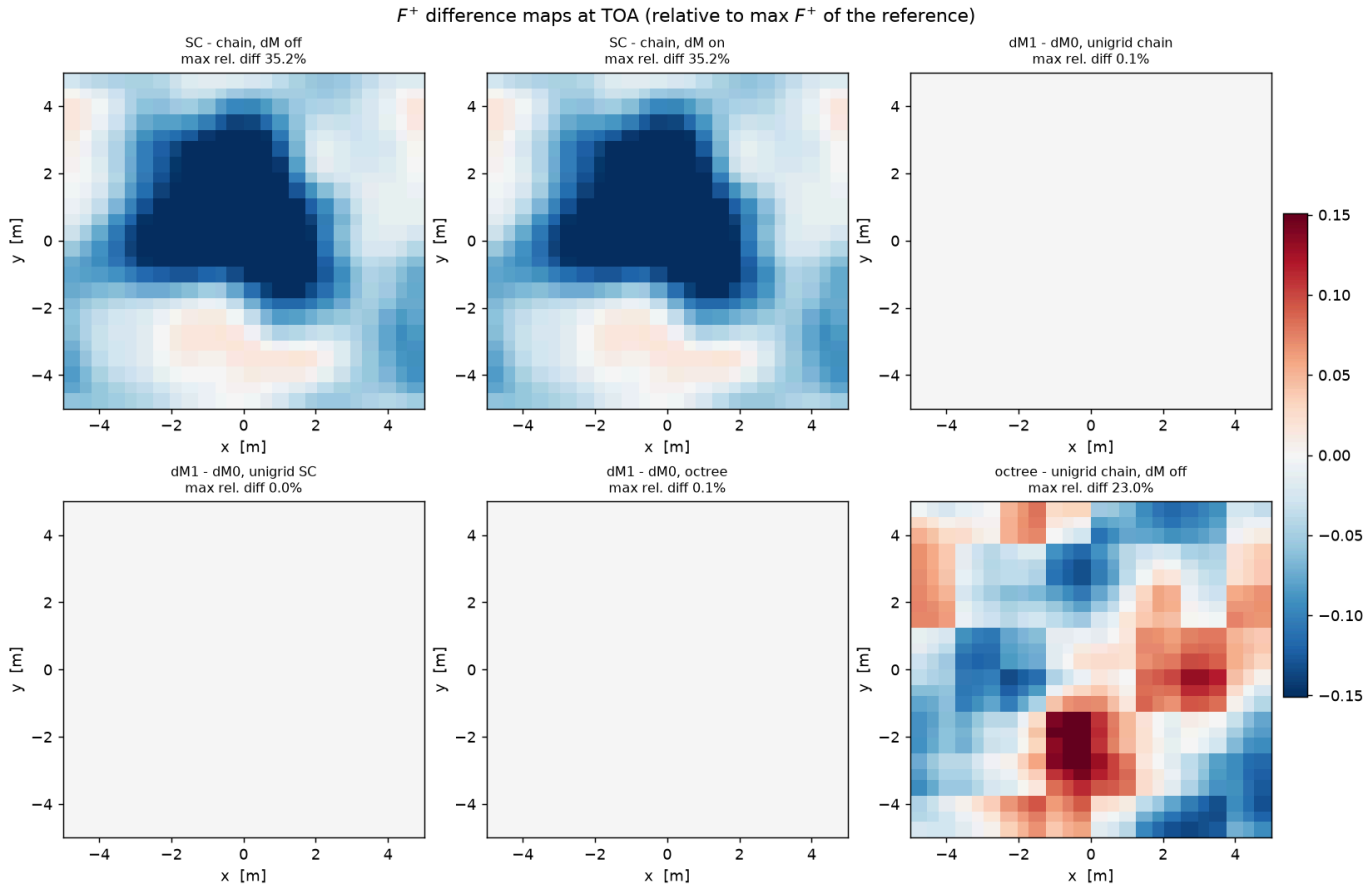


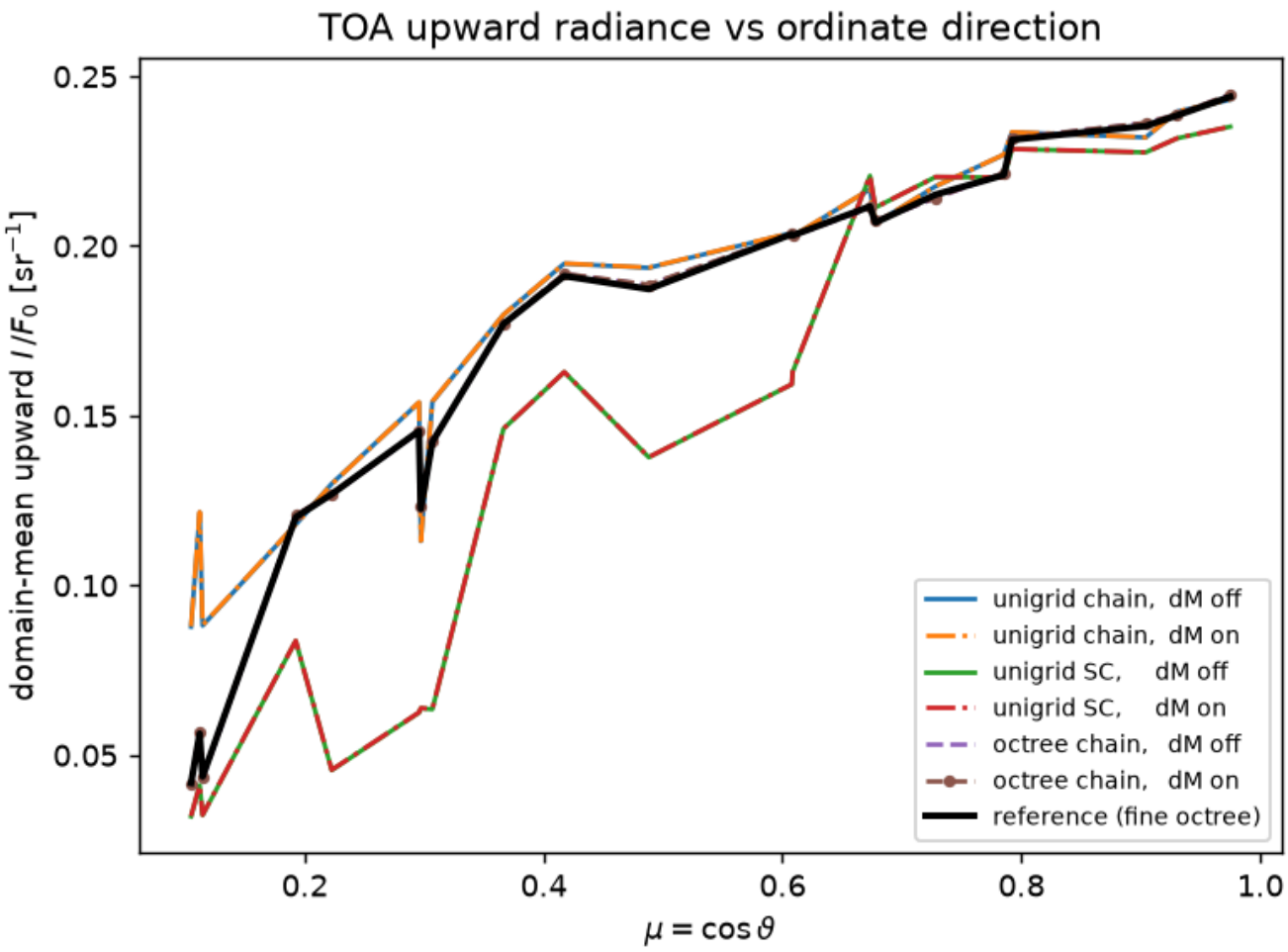
σ^2 is high in the vacuum (collimated beam + background = strongly anisotropic field) and collapses toward zero inside the cloud core where multiple scattering isotropises the field — the "hole" in all six panels. Scheme differences show up mostly in the *shadow region below the cloud* and in the beam path: SC's diffusion

smooths the beam shadow (47.8% max variance difference vs. chain), while the octree resolves the shadow boundary at 0.156 m. delta-M changes σ^2 by ≤0.3%.

4.4 Upward radiance at TOA




F^+ difference maps at TOA (relative to max F^+ of the reference)
SC - chain, dM off
max rel. diff 35.2%
SC - chain, dM on
max rel. diff 35.2%
dM1 - dM0, unigrid chain
max rel. diff 0.1%
dM1 - dM0, unigrid SC
max rel. diff 0.0%
dM1 - dM0, octree
max rel. diff 0.1%
octree - unigrid chain, dM off
max rel. diff 23.0%
x [m]
y [m]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15


TOA upward radiance vs ordinate direction
domain-mean upward I/F_0 [sr^{-1}]
$\mu = \cos\vartheta$
unigrid chain, dM off
unigrid chain, dM on
unigrid SC, dM off
unigrid SC, dM on
octree chain, dM off
octree chain, dM on
reference (fine octree)

- The **chain** runs show a *bright* spot above the cloud (the scattering layer seen from above), the **SC** runs a deep *shadow* blob — the TOA-integrated signature of SC's extra diffusion (max F^+ difference 35%). The octree map is coarse by construction (top-layer cells are 1.25–2.5 m, the cloud is far below and the refinement is κ_e-driven).
- Per-ordinate domain-mean radiance (`toa_mean_radiance.csv`): chain and octree agree closely at all μ; SC collapses at grazing ordinates (μ = 0.10: 0.032 vs 0.088 chain / 0.041 octree) — the SC wrap-ring / grazing-path error — and lies between chain and octree at $\mu \gtrsim 0.7$.
- Domain-mean upward radiance: chain 0.179, octree 0.170, SC 0.142 [F_0 sr^{-1}]; consistently, the TOA net flux is least upward-escaping for SC (mean F_z −2.5935 vs −2.5215/−2.5312).

4.5 Pair statistics (from `pair_stats.csv`)

Relative differences of B − A, normalised by max|A| of the reference run (max over the common lattice / mean over the domain):

pair	J max	J mean	σ^2 max	σ^2 mean	F^+ max	F^+ mean
SC − chain, δ off	31.1%	9.1%	47.8%	6.2%	35.2%	9.8%
SC − chain, δ on	31.1%	9.1%	47.8%	6.2%	35.2%	9.8%
δ1 − δ0, unigrid chain	0.9%	0.01%	0.2%	0.02%	0.1%	0.02%
δ1 − δ0, unigrid SC	0.7%	0.01%	0.2%	0.01%	0.0%	0.02%
δ1 − δ0, octree	1.3%	0.02%	0.3%	0.04%	0.1%	0.05%
octree − unigrid chain, δ off	18.2%	2.8%	23.6%	3.2%	23.0%	7.6%

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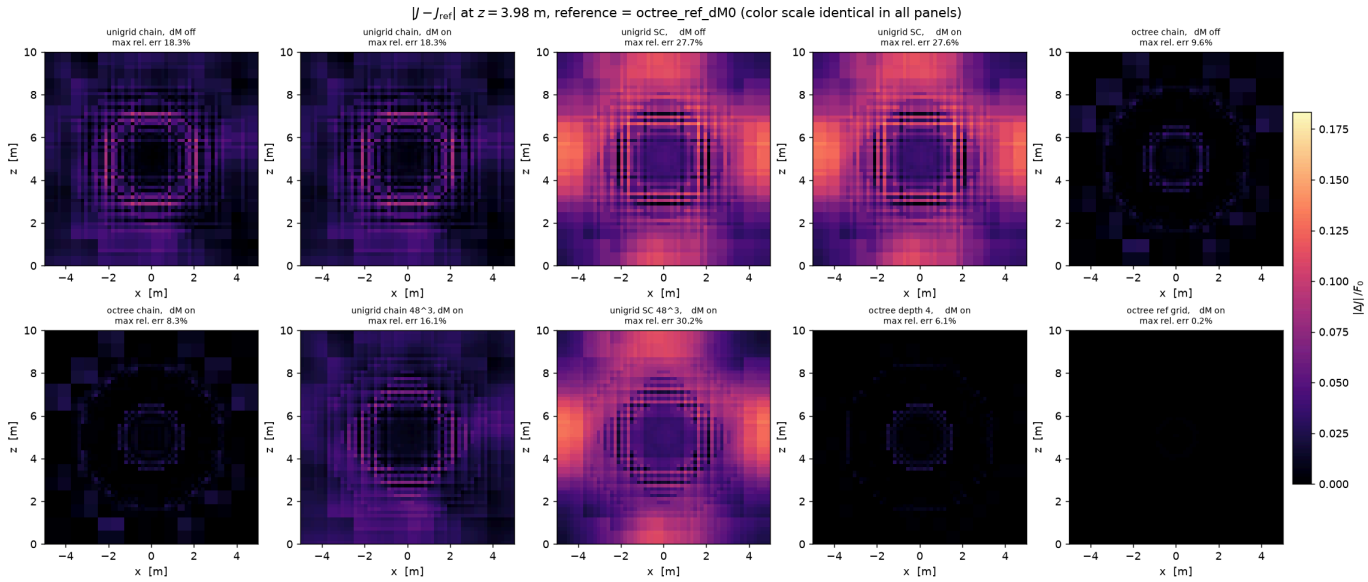
5. Which run is best? — accuracy ranking vs. the reference

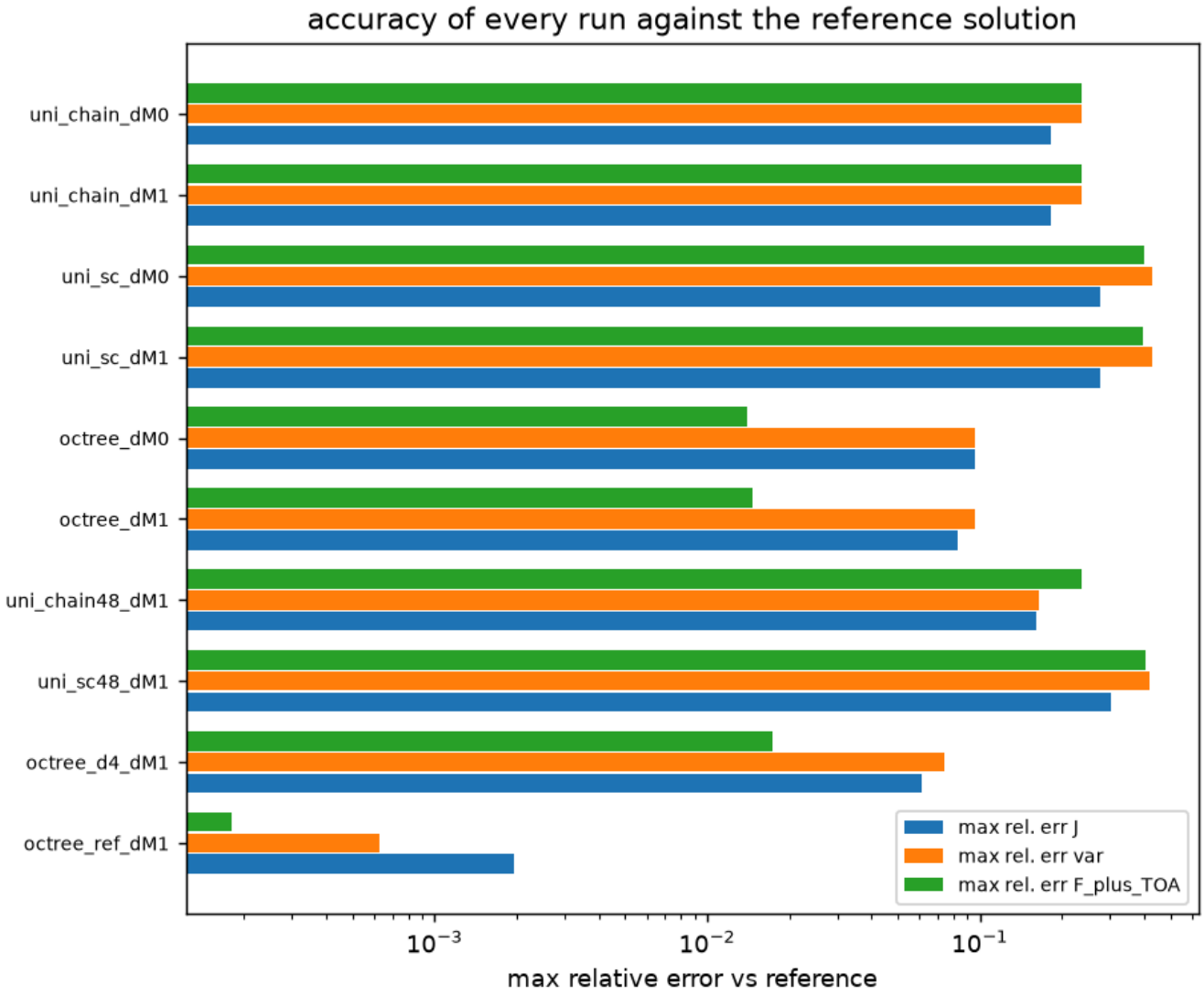
All pairwise numbers above compare approximations *to each other*. Ranking needs a solution that is demonstrably closer to the continuum limit: the fine-octree reference `octree_ref_dM0` (section 2.6). Errors below are relative to max|·| of the reference field on the common lattice (`ranking.csv`, `err_bars.png`, `j_err_maps.png`).

5.1 Ranking

run	wall [s]	J err max	J err mean	σ^2 err max	F^+ err max	ΔF_z @TOA
`octree_ref_dM1`	132.3	0.2%	**0.005%**	0.1%	0.0%	+0.0001
`octree_d4_dM1`	74.5	6.1%	**0.22%**	7.4%	1.7%	+0.0001
`octree_dM1`	47.5	8.3%	**0.47%**	9.6%	1.5%	+0.0011
`octree_dM0`	95.3	9.6%	0.47%	9.6%	1.4%	+0.0008
`uni_chain48_dM1`	19.5	16.1%	**2.7%**	16.4%	23.7%	+0.010
`uni_chain_dM1`	1.5	18.3%	**2.7%**	23.5%	23.5%	+0.010
`uni_chain_dM0`	3.4	18.3%	2.7%	23.6%	23.6%	+0.010
`uni_sc48_dM1`	32.4	30.2%	**6.8%**	41.9%	40.6%	−0.061
`uni_sc_dM1`	7.0	27.6%	**6.9%**	42.8%	39.7%	−0.061
`uni_sc_dM0`	24.7	27.7%	6.9%	42.8%	39.8%	−0.061

(reference `octree_ref_dM0` = 0 by construction; ΔF_z relative to −2.5320: chain +0.4%, SC −2.4%, octree d3 +0.03%)

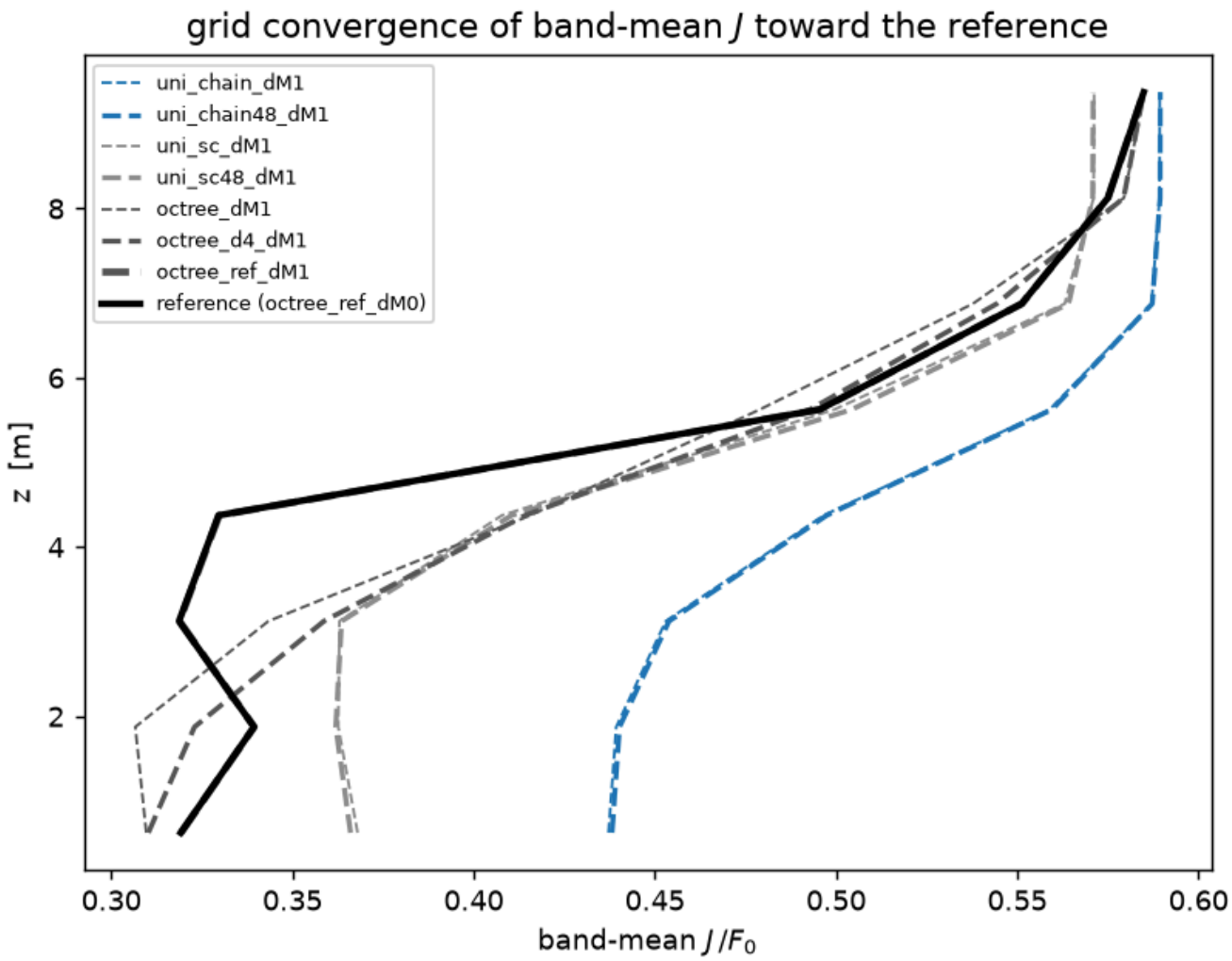


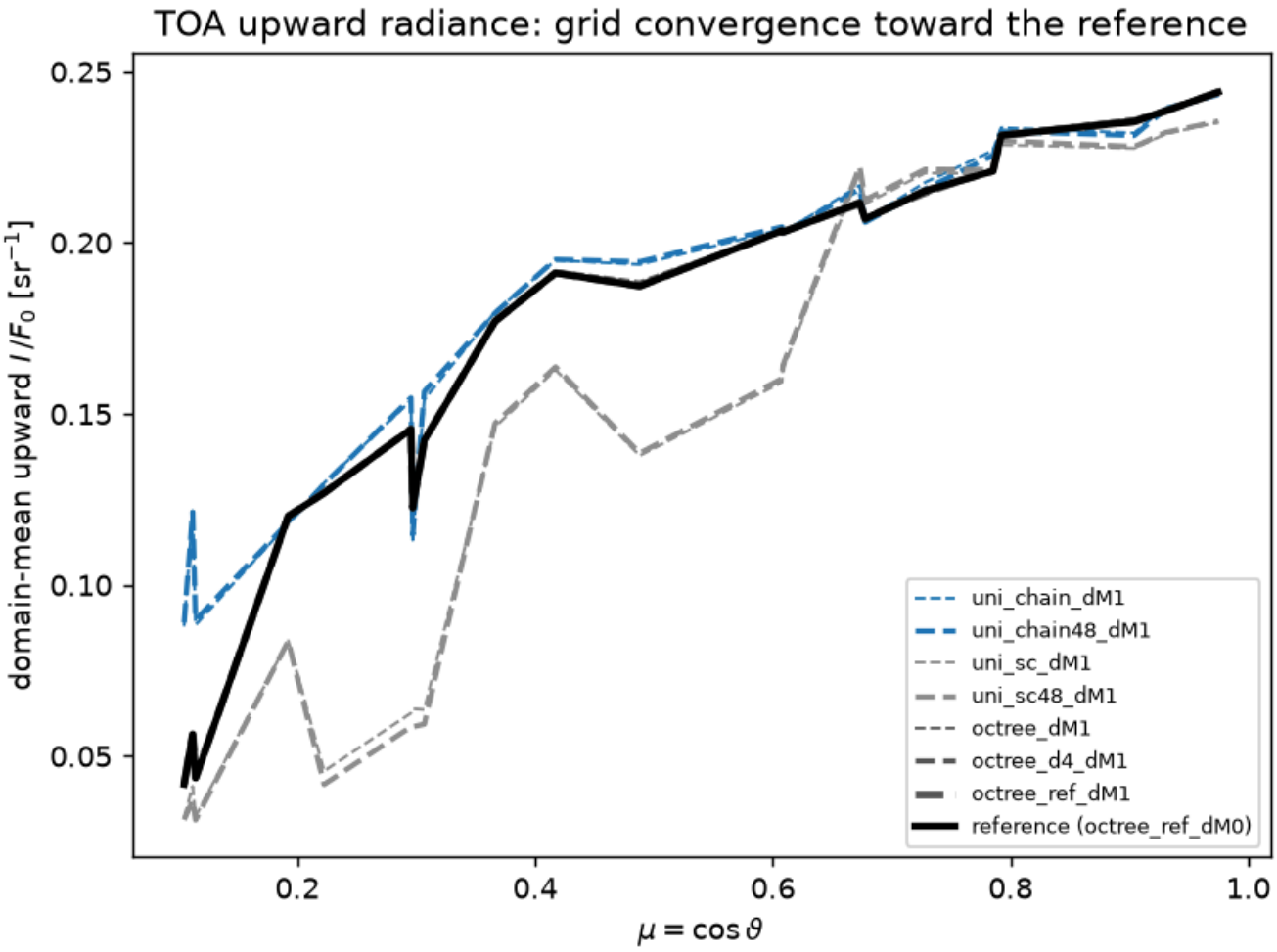


Accuracy order: octree ≫ unigrid chain > unigrid SC. The octree is 6–15× more accurate than chain and 15× more accurate than SC in domain-mean J error; at TOA flux the gap is even larger (0.4% vs 24% chain,

2.4% vs 10% SC mean F^{+} error).

5.2 Convergence evidence (why the ranking is meaningful)





- **Octree** (exact marches, error = cell-size error): mean J error 0.47% (d3) → 0.22% (d4) → reference (d5); max error 9.6% → 6.1% → 0. Roughly halving per refinement level — the octree sequence converges toward the reference.
- **Unigrid chain** 24^3 → 48^3 (8× cells): mean error 2.74% → 2.69%, max 18.3% → 16.1% — *barely moves*. The chain error is dominated by the **grazing-ordinate stencil direction error**: each ordinate is marched along the best *integer lattice direction*, an angular error that does not shrink when h shrinks at fixed 42 ordinates. It converges with N_dir (phase 6), not with h.
- **Unigrid SC** 24^3 → 48^3: mean 6.92% → 6.79%, max 27.7% → 30.2% — also dominated by the angular grazing-ordinate wrap-ring error (section 4.4), not by cell size.

So the unigrid↔reference gap is an *angular path* error of the unigrid transport algorithms at the 42-ordinate development grid; the octree's exact marches remove it, which is why the octree wins by so much and why refining the unigrid alone does not close the gap. (This refines the earlier statement in this report that both errors are "first-order-ish in h": the *cell-size* part is, but on this grid the *stencil-direction* part dominates and is h-independent.)

5.3 Verdict

- **Most accurate** (of the original six): `octree_dM1` — 0.47% mean J error, 1.5% max TOA flux error, 47.5 s. The fine reference grid with δ-M (`octree_ref_dM1`, 132 s) reproduces the δ-off reference to 0.005% mean and is the right choice when the answer itself is the goal.
- **Best accuracy per cost**: `octree_dM1` for physically trustworthy fields; `uni_chain_dM1` (1.5 s, 2.7% mean error) is the cheap workhorse when 3% is acceptable — but its error is angular and will not shrink with

grid refinement at 42 ordinates.

- **delta-M on strictly dominates off** at every grid and scheme: same converged fields (≤1.3% max, ≤0.05% mean), 2–3.5× less wall time. At cloud-core g ≈ 0.95 it should be the default.
- **SC is dominated on both axes**: less accurate *and* more expensive than chain at identical grids (24^3: 6.9% @ 7.0 s vs 2.7% @ 1.5 s).

---

6. Findings

1. **delta-M (M = 1) does exactly what it should**: it cuts the iteration count to 12 for every scheme (from 32/44/30/27), and the converged fields agree with the δ-off runs to ≤1.3% max / ≤0.05% mean — the residual is the known O(τ_cell) explicit-term quadrature gap (it halves to 0.19% at the finest grid), not a physics change. At the cloud-core g ≈ 0.95 it should be the default setting.
2. **Against the fine reference, the accuracy order is octree ≫ chain > SC** (mean J error 0.5% vs 2.7% vs 6.9%). The unigrid↔reference gap is dominated by the *angular* grazing-ordinate errors of both unigrid schemes and does not shrink from 24^3 to 48^3; the octree's exact marches remove it, and the octree error then converges normally with refinement (0.47% → 0.22% → 0).
3. **SC is systematically more diffusive**: lower J inside and below the cloud, deeper TOA shadow, collapsed grazing-ordinate radiance, and a ~2.4% too-small net upward flux at TOA. It is also ~5× slower per iteration than chain on the unigrid (wrap-around ring passes).
4. **The octree buys both resolution and path fidelity**: finer cells where κ_e demands them (0.078 m in the reference cloud core) *and* marching along the true ordinate directions; its exact-march transport removes the chain's grazing-ordinate stencil error, which is why unigrid-chain vs. octree is the manual's anchor-5 pattern.
5. **δ-M is scheme- and grid-agnostic**: the δ-on↔δ-off differences are ~2 orders of magnitude smaller than any scheme comparison, so the comparison is not confounded by the scattering-operator split.
6. **Open point for phase 6**: the ranking is made at fixed 42 ordinates; the common angular error of *all* runs (including the reference) is not captured here. The unigrid schemes' h-independent stencil/wrap errors are expected to shrink with N_dir — the direction-convergence study should re-run this ranking at refinement 2/3.

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7. Interactive 3D scenes

Self-contained vtk.js HTML (open directly in a browser — orbit, actor visibility toggling, colour maps; no server, no Python needed). One scene per run (κ_e isosurfaces at 0.5/1/2/3 m^{-1} + J and σ^2 cut planes at x = 0, y = 0, z = 4) plus difference scenes (|ΔJ| isosurfaces + slices), all regenerated 2026-10-14 on Windows:

```
plot/compare_transport/scene_uni_chain_dM0.html   scene_uni_chain_dM1.html
plot/compare_transport/scene_uni_sc_dM0.html      scene_uni_sc_dM1.html
plot/compare_transport/scene_octree_dM0.html      scene_octree_dM1.html
plot/compare_transport/scene_octree_ref_dM0.html
plot/compare_transport/scene_diff_chain_vs_sc_dM0.html
plot/compare_transport/scene_diff_uni_vs_octree_dM0.html
plot/compare_transport/scene_diff_ref_vs_chain24_dM0.html
plot/compare_transport/scene_diff_ref_vs_sc24_dM0.html
plot/compare_transport/scene_diff_ref_vs_octree3_dM0.html
```

The three `ref_vs_*` scenes are the ranking story made visible: where each coarse run gets J wrong against the reference (chain: below/around the cloud; SC: everywhere diffuse; octree d3: cloud core only).

(pyvista's vtk.js export cannot carry volume mappers, hence slices + isosurfaces; the static volume overview is `vol_j.png`.)

8. Output files

file	content
`summary.csv`	run table (grid, iterations, wall, ⟨J⟩, TOA F_z), all 11 runs
`ranking.csv`	per-run error vs the reference (J, σ^2, F^+, scalars)
`profiles.csv`	band-mean J profiles, all 11 runs
`toa_mean_radiance.csv`	per-ordinate domain-mean upward radiance at TOA
`pair_stats.csv`	max/mean relative differences of all six pairs × J, σ^2, F^+
`j_slices.png`, `j_profiles.png`, `j_diff.png`	mean-intensity figures
`j_err_maps.png`	|J − J_ref| at the cloud plane, every run vs reference
`j_convergence.png`	band-mean J: refinement sequences + reference
`err_bars.png`	max relative error vs reference, all runs (log scale)
`var_slices.png`, `var_diff.png`	angular-variance figures
`toa_fup_maps.png`, `toa_fup_diff.png`, `toa_mu_spectra.png`	TOA radiance figures
`toa_convergence.png`	TOA μ-spectra: refinement sequences + reference
`vol_j.png`	static volume renderings, base six runs + reference
`scene_*.html`	interactive 3D scenes (gitignored, regenerate by rerunning)
`_solve_cache/`	per-run solved fields (gitignored; reruns resume)